

Project 01

Traffic Light Controller

(Timer and Sensor)



RIVOQUIX LEARNING PRIVATE LIMITED
Hustlehub Tech Park,
Somasunderapalya Main Rd, adjacent
27th Main Road, Sector 2 hsr layout,
Karnataka 560102



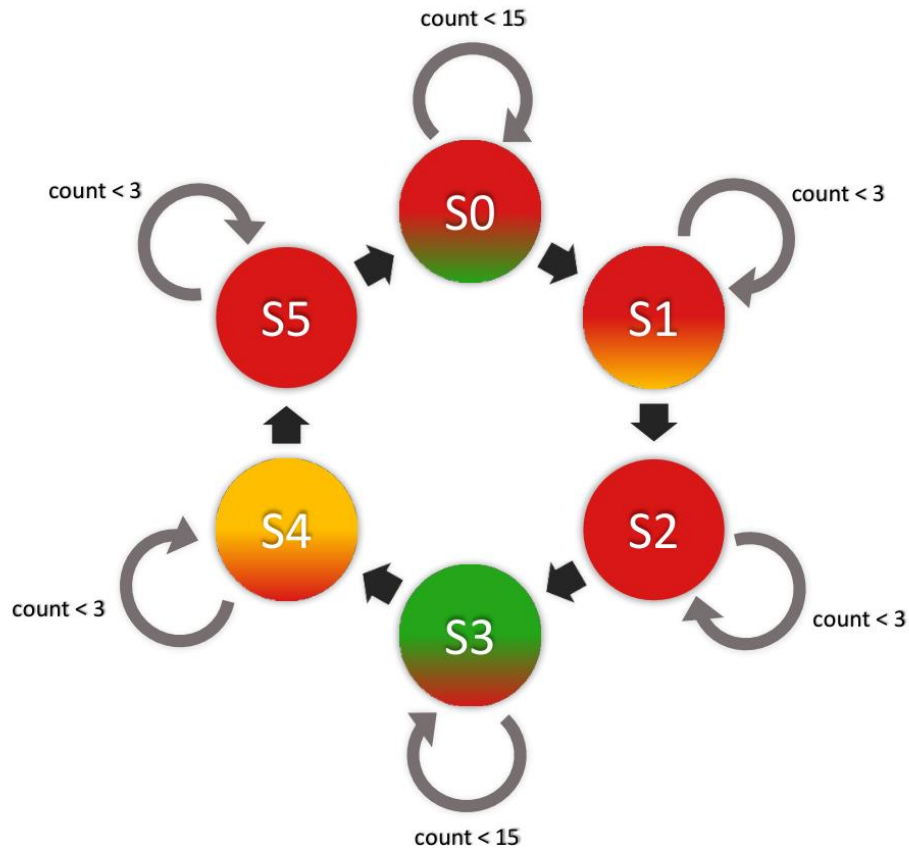
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```

1  `timescale 1ns / 1ps
2
3  module Traffic_Light_Time(
4
5      input wire clk,
6      input wire clr,
7      output reg [5:0] lights
8  );
9
10     reg[2:0] state;
11     reg[3:0] count;
12
13     parameter S0 = 3'b000, S1 = 3'b001, S2 = 3'b010, // states
14               S3 = 3'b011, S4 = 3'b100, S5 = 3'b101;
15
16     parameter SEC5 = 4'b1111, SEC1 = 4'b0011; // delays
17

```



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```
18     always @(posedge clk or posedge clr)
19
20         begin
21             if (clr == 1)
22                 begin
23                     state <= S0;
24                     count <= 0;
25                 end
26             else
```

```
27                 case(state)
28                     S0: if(count < SEC5)
29                         begin
30                             state <= S0;
31                             count <= count + 1;
32                         end
33                     else
34                         begin
35                             state <= S1;
36                             count <= 0;
37                         end
```

```
94     always @(*)
95         begin
96
97             case(state)
98
99                 S0: lights = 6'b100001; // red green
100                S1: lights = 6'b100010; // red yellow
101                S2: lights = 6'b100100; // red red
102                S3: lights = 6'b001100; // green red
103                S4: lights = 6'b010100; // yellow red
104                S5: lights = 6'b100100; // red red
105                default lights = 6'b100001; // red green
106
107             endcase
108         end
109
110 endmodule
```



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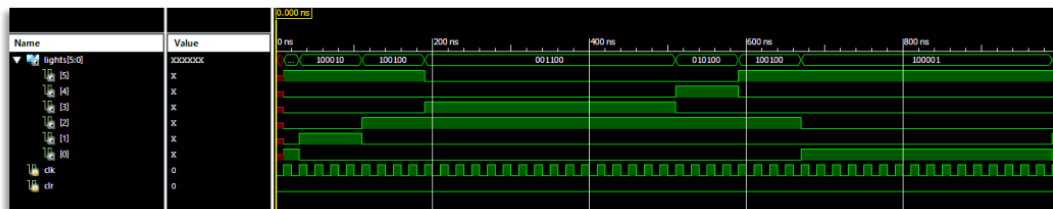


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```

15     initial begin
16
17         clk = 0;
18         clr = 0;
19
20         #100;
21
22     end
23
24     always
25
26     #10 clk = ~clk;
27
28 endmodule

```





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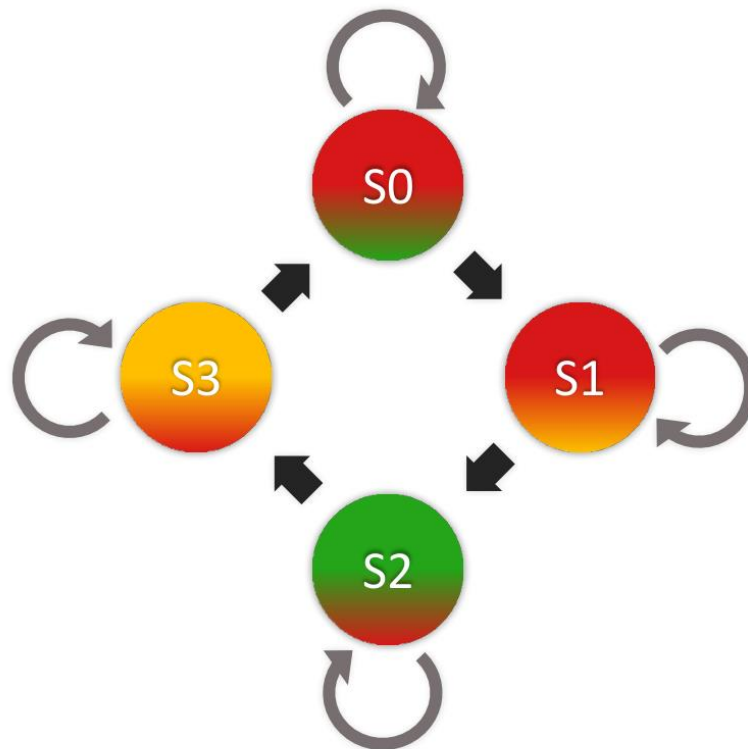
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```

1  `timescale 1ns / 1ps
2
3  module Traffic_Light_Sensor(
4      input clk,
5      input clr,
6      input SNS1,
7      input SNS2,
8      output reg[1:0] TL1, TL2
9  );
10
11  reg [1:0] state, nextstate;
12
13  parameter S0 = 2'b00, S1 = 2'b01, S2 = 2'b10, S3 = 2'b11;
14  parameter green = 2'b00, yellow = 2'b01, red = 2'b10;
15
16
17  always@(posedge clk, posedge clr)
18  begin
19      if(clr) state <= S0;
20      else state <= nextstate;
21  end
  
```



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```
24 always@(*)
25     begin
26         case(state)
27             S0: if(SNS2) nextstate = S1;
28                 else nextstate = S0;
29             S1: nextstate = S2;
30             S2: if(SNS1) nextstate = S3;
31                 else nextstate = S2;
32             S3: nextstate = S0;
33             default: nextstate = S0;
34         endcase
35     end
```

```
38 always@(*)
39     begin
40         if(state == S0)
41             begin
42                 TL1 = green;
43                 TL2 = red;
44             end
45         else if(state == S1)
46             begin
47                 TL1 = yellow;
48                 TL2 = red;
49             end
50         else if(state == S2)
51             begin
52                 TL1 = red;
53                 TL2 = green;
54             end
55         else begin
56             TL1 = red;
57             TL2 = yellow;
58         end
59     end
60 endmodule
```



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```

26     initial
27         begin
28
29             clr = 1;
30             #13
31             clr = 0;
32
33
34             SNS1 = 0; SNS2 = 0; #(10)
35             SNS1 = 0; SNS2 = 1; #(40)
36             SNS1 = 0; SNS2 = 0; #(40)
37             SNS1 = 1; SNS2 = 0; #(30)
38             SNS1 = 0; SNS2 = 0; #(20)
39             SNS1 = 0; SNS2 = 1;
40
41         end
42
43     always
44         begin
45             clk = 1;
46             forever #(10/2) clk = ~clk;
47         end
48
49 endmodule
50

```

